Dear Editors:

We would like to submit the enclosed manuscript entitled “The Sum Is Not the Parts: Generative Artificial Intelligence, Production Networks, and the System-Level Displacement of Youth Employment” for consideration in the Special Issue “Systems Approaches to Generative AI: Workforce Development, Organisational Learning, and Economic Transformation” of Systems. The manuscript has not been published elsewhere and is not under consideration by any other journal. No conflict of interest exists in its submission, and it has been approved for publication by all authors.

Every firm-level study of generative artificial intelligence and youth employment estimates the same object: what happens inside the firm that adopts. This manuscript argues that the object is the wrong one. Firms are not independent units; they buy from one another, sell to one another, and hire from the same local pools of graduates. When one of them adopts a technology that changes what junior labour is for, the consequences travel. We measure where they go. Using 26,115 firm-year observations for 2,800 Chinese A-share listed firms over 2016–2025, their disclosed supplier and customer links, and the 60 local labour markets they share, we estimate a network autoregressive model of the youth employment share and decompose its impacts.

The highlights of the paper are as follows.

(1) Three transmissions, two signs. Own adoption lowers the youth employment share by 0.434 percentage points per standard deviation. Adoption by the firm's customers lowers it by a further 0.396 — 0.94 times the own-firm effect, obtained from firms that may have adopted nothing themselves — and by suppliers by 0.246. But adoption by other firms in the same city raises it, by 0.334: young workers one firm does not hire are hired by the firm next door.

(2) A network multiplier of 3.12. In the network autoregressive model, estimated by generalised spatial two-stage least squares with second-order network lags as excluded instruments, the direct impact is −0.420, the indirect impact −0.893 and the total impact −1.313. That is, 68% of the system-level effect of adoption on youth employment occurs in firms other than the adopter.

(3) Half of the firm-level effect is reallocation, not destruction. Estimated on city aggregates, the same coefficient is −0.222, only 51% of the firm-level estimate. The firm-level coefficient is therefore neither an upper nor a lower bound on anything policy cares about: it overstates the local damage by about a factor of two and understates the system-level damage by about a factor of 3.0. Studies that find large effects and studies that find none may both be right about their own estimand.

(4) The signature is a network, not a common shock. Propagation is concentrated in relationship-specific customer links (−0.530 against −0.260 for generic inputs), is indifferent to whether the customer is in the same city (difference 0.083, $p = 0.584$), is 4.1 times larger for young workers than for experienced ones, and vanishes when the network is randomly rewired while every firm keeps its own adoption and characteristics (randomisation $p = 0.003$).

We believe the manuscript speaks directly to the aims of the Special Issue, which asks for systems approaches to generative AI and workforce development. Our result is a systems result in the strict sense: the effect of the technology on youth employment is a property of the system of firms and not of any firm in it, and it cannot be estimated, or governed, one firm at a time. A young person's chances are being changed by technology decisions taken in firms they will never apply to. Identification rests on the intransitive triads of the disclosed supplier–customer network, which make partners-of-partners adoption an excluded instrument for the endogenous network lag; the instruments pass the Kleibergen–Paap and overidentification tests, the estimates survive a restricted wild cluster bootstrap and coefficient-stability bounds, and every number in every table is produced by the released estimation code. We deeply appreciate your consideration of our manuscript and look forward to receiving comments from the reviewers. Correspondence should be directed to the address below.

Thanks very much for your attention to our paper.

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